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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
Field Sample Permit: Revisionary studies are needed to properly place these, along with previously described taxa at the generic and subtribal levels.
Inclusion and Exclusion Criteria
not required.
Attrition
not required.
Sex as a biological variable
not required.
Subject Demographics
Age: not required.
Weight: not required.
Randomization
Trees were modeled under a Yule process using a random start tree, with the GTR+I+G substitution model for each partition (ITS, combined plastid markers), and a relaxed clock with a lognormal distribution to account for rate heterogeneity among branches.
Blinding
not detected.
Power Analysis
not detected.
Replication
Phylogenetic analyses and divergence time estimation To test for topological congruence, chloroplast and nuclear regions were analyzed separately and phylogenies compared using “tanglegrams” produced in Dendroscope (Scornavacca et al., 2011; Huson and Scornavacca, 2012) as well as by the partition

homogeneity test (ILD) as implemented in PAUP* version 4.0b10 (Swofford, 2002) using 1000 replicates, TBR (tree-bisection reconnection) branch swapping, and MulTrees.
Number: Phylogenetic analyses and divergence time estimation To test for topological congruence, chloroplast and nuclear regions were analyzed separately and phylogenies compared using “tanglegrams” produced in Dendroscope (Scornavacca et al., 2011; Huson and Scornavacca, 2012) as well as by the partition homogeneity test (ILD) as implemented in PAUP* version 4.0b10 (Swofford, 2002) using 1000 replicates, TBR (tree-bisection reconnection) branch swapping, and MulTrees.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and Algorithms			
All PCR reactions were performed with 25 µL of reaction cocktail containing 12.75 µL of sterilized H2O, 2.0 µL of 20 mM dNTPs (Pharmacia Biotech, Piscataway, New Jersey, USA) in an equimolar solution, 2.5 µL of 10x PCR reaction Buffer A (Promega, Madison, Wisconsin, USA), 1.25 µL of 25 mM MgCl2, 0.5 µL 10 mg/ml Bovine Serum Albumin (Sigma-Aldrich, St. Louis, Missouri, USA), 1 µL of 10 mM of each of the two primers, 0.5 Biolase Red Taq DNA polymerase enzyme (Bioline-Thomas Scientific, Swedesboro, New Jersey, USA) and 4 µL of DNA template.	Sigma-Aldrich		
Sequences were edited using Sequencher version 3.1.10 (Gene Codes, Ann Arbor, Michigan, USA).	Sequencher		Suggestion: Sequencher (RRID:SCR_001528)(link)
Alignments were performed using MAFFT version 7.310 (Katoh et al., 2002).	MAFFT		Suggestion: MAFFT (RRID:SCR_011811)(link)
Phylogenetic analyses and divergence time estimation To test for topological congruence, chloroplast and nuclear regions were analyzed separately and phylogenies compared using “tanglegrams” produced in Dendroscope (Scornavacca et al., 2011; Huson and Scornavacca, 2012) as well as by the partition homogeneity test (ILD) as implemented in PAUP* version 4.0b10 (Swofford, 2002) using 1000 replicates, TBR (tree-bisection reconnection) branch swapping, and MulTrees.	PAUP*		Suggestion: PAUP (RRID:SCR_014931)(link)
Maximum likelihood phylogeny inference was done in RAxML version 8.2.12 (Stamatakis, 2014) on the RAxML-HPC2 Workflow on XSEDE on the CIPRES Science Gateway (Miller et al., 2010).	RAxML		
Joint Bayesian inference (BI) on phylogeny and clade divergence times was implemented in BEAST version 2.6.1 (Bouckaert et al., 2014) on the CIPRES Science Gateway (Miller et al., 2010).	BEAST		Suggestion: BEAST (RRID:SCR_010228)(link)

Trees were visualized in FigTree version 1.4.3 (Drummond and Rambaut, 2007; Rambaut, 2017).	FigTree		
Historical biogeography We conducted inference on the biogeographic history of the Vernonieae using the maximum likelihood implementation of the program BioGeoBEARS version 1.1.2 (Matzke, 2013a, b, 2014; Dupin et al., 2016) Ancestral ranges and biogeographic events were inferred using the Bayesian calibrated tree trimmed to include only members of the Vernonieae tribe.	BioGeoBEARS		

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For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq